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Monash University

Semester Two Examination Period 2013

Faculty of Engineering

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EXAM CODES: **MAE3405**

TITLE OF PAPER: **Flight Vehicle Propulsion**

EXAM DURATION: 3 hours writing time

READING TIME: 10 minutes

THIS PAPER IS FOR STUDENTS STUDYING AT:(tick where applicable)

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☐ Caulfield ☐ Gippsland ☐ Peninsula ☐ Enhancement Studies ☐ Sth Africa
☐ Pharmacy ☐ Other (specify)

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Note the following: To gain full marks you must: (a) correctly complete all three questions; (b) list assumptions used, provide the necessary diagrams and show all steps in your solutions; and (c) if you use an equation that does not appear on the equation sheet, show how the equation can be derived from those provided, or give a simple derivation of the equation, or show how it can be determined from the relevant thermodynamic cycle.

1. An aircraft is flying steady and level at 2000m with an equivalent airspeed of 150Knots. It is powered by a 6 cylinder, 4-stroke spark ignition piston engine whose data is tabulated below. Assuming the engine to operate on the fuel/air Otto cycle and the fuel to be Avgas, determine: **(a)** the drag acting on the aircraft assuming the propeller to have a downwash velocity equal to 17.5% of the true airspeed; **(b)** the fuel consumption in litres/hour. [22+3=25 Marks]

Quantity	**Value**	**Quantity**	**Value**
Compression ratio	7	Combustion efficiency	90%
Bore	0.1m	Mechanical Efficiency	85%
Stroke	0.12m	Volumetric efficiency	80%
RPM	2700	Air to fuel ratio	14.7
Cp reactants	1.055kJ/kg-K	Cp products	1.486kJ/kg-K
Reactant mole weight	28.9kmol/kg	Product mole weight	28.0kmol/kg

2. A helicopter blade has a chord of 250mm and its tip is 3.5m from its centre of rotation. The blade is fixed a distance of 0.4m from the centre of rotation. A shaft from the engine drives the blade. The angle of attack of the blade is such that the lift coefficient is constant at C_l=1.0 along its length. The advance speed can be assumed to be zero. **(a)** If the blade mass is 20kg, at what RPM will the drive shaft experience zero vertical force at SSL? **(b)** If there are 4 blades attached to the shaft, estimate the power required to drive the rotor. [10+5=15 Marks]

3. A new twin spool, aft-fan engine is being designed and you are tasked with calculating the cruise design point thrust (Ma=0.8 at 10,000m). The layout of this engine is based on a gas generator spool and a free (or power) spool arrangement. For the gas generator spool, the turbine stage drives the compressor stage, while for the free spool, which is located directly after the gas generator, the turbine directly drives the aft-fan. Similar to a turbofan engine, the fan on this engine is housed in a duct which passes the cold flow to a nozzle. Similar to a turboprop, the hot flow is passed to a nozzle with a fixed nozzle pressure ratio (PNPR), see below. Being an aft-fan engine, there are separate inlet nozzles for the engine core and the fan. [30 Marks]

Quantity	**Value**	**Quantity**	**Value**
CPR	15	Combustor pressure drop	3%
PNPR	1.2	Compressor Cp	1.005kJ/kg-K
Combustor F/A ratio	0.02	Compressor γ	1.4
Isentropic Efficiencies		Turbine Cp	1.148kJ/kg-K
- Fan & Compressor	85%	Turbine γ	1.333
- Turbines	90%	Exhaust gas constant R	0.287kJ/kg-K
- Intakes	95%	Fan Intake nozzle area	2.5m^2
- Propelling nozzles	95%	Core Intake nozzle area	0.5m^2
Generator spool eff.	99%	Fuel	Jet A1
Combustor efficiency	98%		

Standard Atmosphere

Altitude Km	Temperature K	Pressure Pa	Density kg/m^3	Speed of Sound m/s
0	288.15	101325	1.225	340.29
1	281.65	89874.5	1.111	336.43
2	275.15	79495.2	1.006	332.52
3	268.65	70108.5	0.9091	328.57
4	262.15	61640.2	0.8191	324.57
5	255.65	54019.9	0.7361	320.52
6	249.15	47181.0	0.6597	316.42
7	242.65	41060.7	0.5895	312.27
8	236.15	35599.8	0.5252	308.06
9	229.65	30742.4	0.4663	303.79
10	223.15	26436.2	0.4127	299.46
11	216.65	22632.0	0.3639	295.06
12	216.65	19330.4	0.3108	295.06
13	216.65	16510.4	0.2655	295.06
14	216.65	14101.7	0.2268	295.06
15	216.65	12044.5	0.1937	295.06
16	216.65	10287.4	0.1654	295.06
17	216.65	8786.68	0.1413	295.06
18	216.65	7504.84	0.1207	295.06
19	216.65	6410.00	0.1031	295.06
20	216.65	5474.88	0.088	295.06

Gases Properties

Gas	M_{wt} kg/kmol	R J/kg-K	Cp J/kg-K	Cv J/kg-K	γ	h_f° MJ/kmole
Air (O2+3.76N2)	29	287.00	1005.0	718.0	1.4	0
Oxygen	32	259.7	921.6	661.9	1.4	0
Carbon dioxide	44	188.92	841.8	652.9	1.289	-393.5
Nitrogen	28	296.80	1041.6	744.8	1.4	0
Water (gas)	18	461.52	1872.3	1410.8	1.327	-241.8

Liquid Fuel Property Data

Fuel	Density kg/m^3	LHV MJ/kg	LHV MJ/litre
Avgas	708.2	45.0	31.9
Jet A1	793.0	43.1	34.2

Planetary Data for the Earth

Quantity	Value
Gravity	$9.81 m/s^2$
Mean Gas Constant	287J/kg-K

Conversion and Reference Data

1Knot = 1.85kph Reference Temperature 298K R_u=8.314 kJ/kmol.K

***Equation Sheet:* state condition numbering used in equations is arbitrary.**

$$S = bc \qquad A = \frac{b}{c} \qquad \mathrm{V_{cal}} = \mathrm{V_E} = \mathrm{V_0}\sqrt{\frac{\rho}{\rho_s}} = \mathrm{V_0}\sqrt{\sigma}$$

$$L = \frac{1}{2}\rho U^2 c_L S \qquad D = \frac{1}{2}\rho U^2 c_D S \qquad c_L = a\alpha \qquad \eta_p = \frac{1}{1 + w/V}$$

$$\dot{W} = (IMEP)\dot{V}_s \qquad \dot{V}_s = A_P L_S \frac{N}{2} n \qquad \eta = \frac{\dot{W}}{\dot{Q}_H} \qquad \dot{m} = \rho V A$$

$$TSFC = \frac{\dot{m}_f}{F_N} \qquad BSFC = \frac{\dot{m}_f}{\dot{W}_b} \qquad F_s = \frac{F_N}{\dot{m}_a} \qquad f = \frac{\dot{m}_f}{\dot{m}_a}$$

$$\dot{W}_b = T(w + V) \qquad w = \frac{1}{2}\left[\left(V^2 + \frac{2T}{\rho A_p}\right)^{1/2} - V\right] \qquad \dot{W}_{b,o} = T_o^{3/2}/\sqrt{2\rho A_p}$$

$$p = \rho RT \qquad pv = RT \qquad \gamma = C_p/C_v \qquad R = C_p - C_v$$

$$(pv^\gamma)_1 = (pv^\gamma)_2 \qquad (Tv^{\gamma-1})_1 = (Tv^{\gamma-1})_2$$

$$\sum Q + \sum W = 0 \qquad Q + W = \Delta U \qquad LHV = \left[(h_f^o)_R - (h_f^o)_P\right]/M_{wf}$$

$$\left(\frac{p}{\rho} + u + \frac{C^2}{2}\right)_1 = \left(\frac{p}{\rho} + u + \frac{C^2}{2}\right)_2 + Q + W$$

$$\left(\frac{p}{\rho} + u\right) = h \qquad (h_2 - h_1) = C_p(T_2 - T_1) \qquad h = C_p(T - T_R)$$

$$\frac{p_0}{p} = \left(\frac{T_0}{T}\right)^{\frac{\gamma}{\gamma-1}} \qquad \frac{p_{02}}{p_{01}} = \left(\frac{T_{02'}}{T_{01}}\right)^{\frac{\gamma}{\gamma-1}} \qquad \eta_i = \frac{T_{01'} - T_a}{T_{01} - T_a}$$

$$T_{0a} = T_a + \frac{C_a^2}{2c_p} \qquad \frac{p_{01}}{p_a} = \left[1 + \eta_i \frac{C_a^2}{2c_p T_a}\right]^{\frac{\gamma}{\gamma-1}} \qquad TP = \dot{W}_s \eta_p + C_a F_N \qquad \eta_c = \frac{T_{02'} - T_{01}}{T_{02} - T_{01}}$$

$$\eta_t = \frac{T_{03} - T_{04}}{T_{03} - T_{04'}} \qquad \eta_j = \frac{T_{04} - T_5}{T_{04} - T_{5'}} \qquad \frac{p_{04}}{p_c} = \left[1 - \frac{1}{\eta_j}\left(\frac{\gamma - 1}{\gamma + 1}\right)\right]^{-\left(\frac{\gamma}{\gamma-1}\right)} \qquad \frac{T_c}{T_{04}} = \left(\frac{2}{\gamma + 1}\right)$$

$$a = \sqrt{R\gamma T_c} \qquad M = \frac{C}{a} \qquad B = \frac{\dot{m}_c}{\dot{m}_h} \qquad FPR = p_{02}/p_{01} \quad OPR = p_{03}/p_{01}$$

$$C_J = \left[\left(\frac{2\gamma}{\gamma-1}\right)RT_c\left(1-\left(\frac{p_e}{p_c}\right)^{(\gamma-1)/\gamma}\right)\right]^{1/2} \qquad \frac{A}{A^*} = \frac{1}{M}\left(\frac{2+(\gamma-1)M^2}{\gamma+1}\right)^{(\gamma+1)/(2\gamma-2)}$$

$$\Delta C = C_e \ln(M_R) - gt_b \sin\theta \qquad z_b = C_o t_o + C_e t_b\left(\frac{1}{M_R - 1}\ln\left(\frac{1}{M_R}\right)+1\right) - \frac{gt_b^2 \sin\theta}{2}$$

$$z_c = z_b + \frac{r_o C_b^2}{2g_o r_o - C_b^2} \qquad M_R = \frac{M_o}{M_f} \qquad C_{esc} = r_o\sqrt{\left(\frac{2g_o}{r_o+z}\right)} \qquad C_{sat} = r_o\sqrt{\left(\frac{g_o}{r_o+z}\right)}$$

Normal Shock Tables

TABLE E.4 Normal Shock Flow of an Ideal Gas (k = 1.4)

Ma_x	Ma_y	$\frac{p_x}{p_x}$	T_y/T_x	$\rho_y/\rho_x = V_x/V_y$	$\frac{p_{0,y}}{p_{0,x}}$	$p_{0,y}/p_x$	Ma_x	Ma_y	$\frac{p_y}{p_x}$	T_y/T_x	$\rho_y/\rho_x = V_x/V_y$	$\frac{p_{0,y}}{p_{0,x}}$	$p_{0,y}/p_x$
1.00	1.00000	1.00000	1.00000	1.00000	1.00000	1.8929	1.49	0.70466	2.4234	1.3136	1.8449	0.93292	3.3756
1.01	0.99013	1.02345	1.00665	1.01669	0.99999	1.9152	1.50	0.70109	2.4583	1.3202	1.8621	0.92978	3.4133
1.02	0.98052	1.04713	1.01325	1.03344	0.99998	1.9379	1.51	0.69758	2.4934	1.3269	1.8792	0.92658	3.4512
1.03	0.97115	1.07105	1.01981	1.05024	0.99997	1.9610	1.52	0.69413	2.5288	1.3336	1.8962	0.92331	3.4894
1.04	0.96202	1.09520	1.02634	1.06709	0.99994	1.9845	1.53	0.69073	2.5644	1.3403	1.9133	0.91999	3.5279
1.05	0.95312	1.1196	1.03284	1.08398	0.99987	2.0083	1.54	0.68739	2.6003	1.3470	1.9303	0.91662	3.5667
1.06	0.94444	1.1442	1.03931	1.10092	0.99976	2.0325	1.55	0.68410	2.6363	1.3538	1.9473	0.91319	3.6058
1.07	0.93598	1.1690	1.04575	1.11790	0.99962	2.0570	1.56	0.68086	2.6725	1.3606	1.9643	0.90970	3.6451
1.08	0.92772	1.1941	1.05217	1.13492	0.99944	2.0819	1.57	0.67768	2.7090	1.3674	1.9812	0.90615	3.6847
1.09	0.91965	1.2194	1.05856	1.15199	0.99921	2.1072	1.58	0.67455	2.7458	1.3742	1.9981	0.90255	3.7245
1.10	0.91177	1.2450	1.06494	1.1691	0.99892	2.1328	1.59	0.67147	2.7828	1.3811	2.0149	0.89889	3.7645
1.11	0.90408	1.2708	1.07130	1.1862	0.99858	2.1588	1.60	0.66844	2.8201	1.3880	2.0317	0.89520	3.8049
1.12	0.89656	1.2968	1.07764	1.2034	0.99820	2.1851	1.61	0.66545	2.8575	1.3949	2.0485	0.89144	3.8456
1.13	0.88922	1.3230	1.08396	1.2206	0.99776	2.2118	1.62	0.66251	2.8951	1.4018	2.0652	0.88764	3.8866
1.14	0.88204	1.3495	1.09027	1.2378	0.99726	2.2388	1.63	0.65962	2.9330	1.4088	2.0820	0.88380	3.9278
1.15	0.87502	1.3762	1.09657	1.2550	0.99669	2.2661	1.64	0.65677	2.9712	1.4152	2.0986	0.87992	3.9693
1.16	0.86816	1.4032	1.10287	1.2723	0.99605	2.2937	1.65	0.65396	3.0096	1.4228	2.1152	0.87598	4.0111
1.17	0.86145	1.4304	1.10916	1.2896	0.99534	2.3217	1.66	0.65119	3.0482	1.4298	2.1318	0.87201	4.0531
1.18	0.85488	1.4578	1.11544	1.3069	0.99455	2.3499	1.67	0.64847	3.0870	1.4369	2.1484	0.86800	4.0954
1.19	0.84846	1.4854	1.12172	1.3243	0.99371	2.3786	1.68	0.64579	3.1261	1.4440	2.1649	0.86396	4.1379
1.20	0.84217	1.5133	1.1280	1.3416	0.99280	2.4075	1.69	0.64315	3.1654	1.4512	2.1813	0.85987	4.1807
1.21	0.83601	1.5414	1.1343	1.3590	0.99180	2.4367	1.70	0.64055	3.2050	1.4583	2.1977	0.85573	4.2238
1.22	0.82998	1.5698	1.1405	1.3764	0.99073	2.4662	1.71	0.63798	3.2448	1.4655	2.2141	0.85155	4.2672
1.23	0.82408	1.5984	1.1468	1.3938	0.98975	2.4961	1.72	0.63545	3.2848	1.4727	2.2304	0.84735	4.3108
1.24	0.81830	1.6272	1.1531	1.4112	0.98835	2.5263	1.73	0.63296	3.3250	1.4800	2.2467	0.84312	4.3547
1.25	0.81264	1.6562	1.1594	1.4286	0.98706	2.5568	1.74	0.63051	3.3655	1.4873	2.2629	0.83886	4.3989
1.26	0.80709	1.6855	1.1657	1.4460	0.98568	2.5876	1.75	0.62809	3.4062	1.4946	2.2791	0.83456	4.4433
1.27	0.80165	1.7150	1.1720	1.4634	0.98422	2.6187	1.76	0.62570	3.4472	1.5019	2.2952	0.83024	4.4880
1.28	0.79631	1.7448	1.1782	1.4808	0.98268	2.6500	1.77	0.62335	3.4884	1.5093	2.3113	0.82589	4.5330
1.29	0.79108	1.7748	1.1846	1.4983	0.98106	2.6816	1.78	0.62104	3.5298	1.5167	2.3273	0.82152	4.5783
1.30	0.78596	1.8050	1.1909	1.5157	0.97935	2.7135	1.79	0.61875	3.5714	1.5241	2.3433	0.81711	4.6238
1.31	0.78093	1.8354	1.1972	1.5331	0.97758	2.7457	1.80	0.61650	3.6133	1.5316	2.3592	0.81268	4.6695
1.32	0.77600	1.8661	1.2035	1.5505	0.97574	2.7783	1.81	0.61428	3.6554	1.5391	2.3751	0.80823	4.7155
1.33	0.77116	1.8970	1.2099	1.5680	0.97382	2.8112	1.82	0.61209	3.6978	1.5466	2.3909	0.80376	4.7618
1.34	0.76641	1.9282	1.2162	1.5854	0.97181	2.8444	1.83	0.60993	3.7404	1.5542	2.4067	0.79926	4.8083
1.35	0.76175	1.9596	1.2226	1.6028	0.96972	2.8778	1.84	0.60780	3.7832	1.5617	2.4224	0.79474	4.8551
1.36	0.75718	1.9912	1.2290	1.6202	0.96756	2.9115	1.85	0.60570	3.8262	1.5694	2.4381	0.79021	4.9022
1.37	0.75269	2.0230	1.2354	1.6376	0.96534	2.9455	1.86	0.60363	3.8695	1.5770	2.4537	0.78567	4.9498
1.38	0.74828	2.0551	1.2418	1.6550	0.96304	2.9798	1.87	0.60159	3.9130	1.5847	2.4693	0.78112	4.9974
1.39	0.74396	2.0874	1.2482	1.6723	0.96065	3.0144	1.88	0.59957	3.9568	1.5924	2.4848	0.77656	5.0453
1.40	0.73971	2.1200	1.2547	1.6896	0.95819	3.0493	1.89	0.59758	4.0008	1.6001	2.5003	0.77197	5.0934
1.41	0.73554	2.1528	1.2612	1.7070	0.95566	3.0844	1.90	0.59562	4.0450	1.6079	2.5157	0.76735	5.1417
1.42	0.73144	2.1858	1.2676	1.7243	0.95306	3.1198	1.91	0.59368	4.0894	1.6157	2.5310	0.76273	5.1904
1.43	0.72741	2.2190	1.2742	1.7416	0.95039	3.1555	1.92	0.59177	4.1341	1.6236	2.5463	0.75812	5.2394
1.44	0.72345	2.2525	1.2807	1.7589	0.94765	3.1915	1.93	0.58988	4.1790	1.6314	2.5615	0.75347	5.2886
1.45	0.71956	2.2862	1.2872	1.7761	0.94483	3.2278	1.94	0.58802	4.2242	1.6394	2.5767	0.74883	5.3381
1.46	0.71574	2.3202	1.2938	1.7934	0.94196	3.2643	1.95	0.58618	4.2696	1.6473	2.5919	0.74418	5.3878
1.47	0.71198	2.3544	1.3004	1.8106	0.93901	3.3011	1.96	0.58437	4.3152	1.6553	2.6070	0.73954	5.4378
1.48	0.70829	2.3888	1.3070	1.8278	0.93600	3.3382	1.97	0.58258	4.3610	1.6633	2.6220	0.73487	5.4880

Normal Shock Tables Continued

TABLE E.4 (continued)

Ma_x	Ma_y	$\frac{p_y}{p_x}$	T_y/T_x	$\rho_y/\rho_x = V_x/V_y$	$\frac{p_{0,y}}{p_{0,x}}$	$p_{0,y}/p_x$	Ma_x	Ma_y	$\frac{p_y}{p_x}$	T_y/T_x	$\rho_y/\rho_x = V_x/V_y$	$\frac{p_{0,y}}{p_{0,x}}$	$p_{0,y}/p_x$
1.98	0.58081	4.4071	1.6713	2.6369	0.73021	5.5385	2.55	0.50831	7.4196	2.1875	3.3918	0.47927	8.8505
1.99	0.57907	4.4534	1.6794	2.6518	0.72554	5.5894	2.56	0.50740	7.4792	2.1976	3.4034	0.47540	8.9162
2.00	0.57735	4.5000	1.6875	2.6666	0.72088	5.6405	2.57	0.50651	7.5391	2.2077	3.4149	0.47155	8.9821
2.01	0.57565	4.5468	1.6956	2.6814	0.71619	5.6918	2.58	0.50562	7.5992	2.2179	3.4263	0.46772	9.0482
2.02	0.57397	4.5938	1.7038	2.6962	0.71152	5.7434	2.59	0.50474	7.6595	2.2281	3.4376	0.46391	9.1146
2.03	0.57231	4.6411	1.7120	2.7109	0.70686	5.7952	2.60	0.50387	7.7200	2.2383	3.4489	0.46012	9.1813
2.04	0.57068	4.6886	1.7203	2.7255	0.70218	5.8473	2.61	0.50301	7.7808	2.2486	3.4602	0.45636	9.2481
2.05	0.56907	4.7363	1.7286	2.7400	0.69752	5.8997	2.62	0.50216	7.8418	2.2589	3.4714	0.45262	9.3154
2.06	0.56747	4.7842	1.7369	2.7545	0.69284	5.9523	2.63	0.50132	7.9030	2.2693	3.4825	0.44891	9.3829
2.07	0.56589	4.8324	1.7452	2.7690	0.68817	6.0052	2.64	0.50048	7.9645	2.2797	3.4936	0.44522	9.4507
2.08	0.56433	4.8808	1.7536	2.7834	0.68351	6.0584	2.65	0.49965	8.0262	2.2901	3.5047	0.44155	9.5187
2.09	0.56280	4.9295	1.7620	2.7977	0.67886	6.1118	2.66	0.49883	8.0882	2.3006	3.5157	0.43791	9.5869
2.10	0.56128	4.9784	1.7704	2.8119	0.67422	6.1655	2.67	0.49802	8.1504	2.3111	3.5266	0.43429	9.6553
2.11	0.55987	5.0275	1.7789	2.8216	0.66957	6.2194	2.68	0.49722	8.2128	2.3217	3.5374	0.43070	9.7241
2.12	0.55830	5.0768	1.7874	2.8402	0.66492	6.2736	2.69	0.49642	8.2754	2.3323	3.5482	0.42713	9.7932
2.13	0.55683	5.1264	1.7960	2.8543	0.66029	6.3280	2.70	0.49563	8.3383	2.3429	3.5590	0.42359	9.8625
2.14	0.55538	5.1762	1.8046	2.8683	0.65567	6.3827	2.71	0.49485	8.4014	2.3536	3.5697	0.42007	9.9320
2.15	0.55395	5.2262	1.8132	2.8823	0.65105	6.4377	2.72	0.49408	8.4648	2.3643	3.5803	0.41657	10.002
2.16	0.55254	5.2765	1.8219	2.8962	0.64644	6.4929	2.73	0.49332	8.5284	2.3750	3.5909	0.41310	10.072
2.17	0.55114	5.3270	1.8306	2.9100	0.64185	6.5484	2.74	0.49256	8.5922	2.3858	3.6014	0.40965	10.142
2.18	0.54976	5.3778	1.8393	2.9238	0.63728	6.6042	2.75	0.49181	8.6562	2.3966	3.6119	0.40622	10.212
2.19	0.54841	5.4288	1.8481	2.9376	0.63270	6.6602	2.76	0.49107	8.7205	2.4074	3.6224	0.40282	10.283
2.20	0.54706	5.4800	1.8569	2.9512	0.62812	6.7163	2.77	0.49033	8.7850	2.4183	3.6328	0.39945	10.354
2.21	0.54572	5.5314	1.8657	2.9648	0.62358	6.7730	2.78	0.48960	8.8497	2.4292	3.6431	0.39610	10.426
2.22	0.54440	5.5831	1.8746	2.9783	0.61905	6.8299	2.79	0.48888	8.9147	2.4402	3.6533	0.39276	10.498
2.23	0.54310	5.6350	1.8835	2.9918	0.61453	6.8869	2.80	0.48817	8.9800	2.4512	3.6635	0.38946	10.569
2.24	0.54182	5.6872	1.8924	3.0052	0.61002	6.9442	2.81	0.48746	9.0454	2.4622	3.6737	0.38618	10.641
2.25	0.54055	5.7396	1.9014	3.0186	0.60554	7.0018	2.82	0.48676	9.1111	2.4733	3.6838	0.38293	10.714
2.26	0.53929	5.7922	1.9104	3.0319	0.60106	7.0597	2.83	0.48607	9.1770	2.4844	3.6939	0.37970	10.787
2.27	0.53805	5.8451	1.9194	3.0452	0.59659	7.1178	2.84	0.48538	9.2432	2.4955	3.7039	0.37649	10.860
2.28	0.53683	5.8982	1.9285	3.0584	0.59214	7.1762	2.85	0.48470	9.3096	2.5067	3.7139	0.37330	10.933
2.29	0.53561	5.9515	1.9376	3.0715	0.58772	7.2348	2.86	0.48402	9.3762	2.5179	3.7238	0.37013	11.006
2.30	0.53441	6.0050	1.9468	3.0846	0.58331	7.2937	2.87	0.48334	9.4431	2.5292	3.7336	0.36700	11.080
2.31	0.53322	6.0588	1.9560	3.0976	0.57891	7.3529	2.88	0.48268	9.5102	2.5405	3.7434	0.36389	11.154
2.32	0.53205	6.1128	1.9652	3.1105	0.57452	7.4123	2.89	0.48203	9.5775	2.5518	3.7532	0.36080	11.228
2.33	0.53089	6.1670	1.9745	3.1234	0.57015	7.4720	2.90	0.48138	9.6450	2.5632	3.7629	0.35773	11.302
2.34	0.52974	6.2215	1.9838	3.1362	0.56580	7.5319	2.91	0.48074	9.7127	2.5746	3.7725	0.35469	11.377
2.35	0.52861	6.2762	1.9931	3.1490	0.56148	7.5920	2.92	0.48010	9.7808	2.5860	3.7821	0.35167	11.452
2.36	0.52749	6.3312	2.0025	3.1617	0.55717	7.6524	2.93	0.47946	9.8491	2.5975	3.7917	0.34867	11.527
2.37	0.52638	6.3864	2.0119	3.1743	0.55288	7.7131	2.94	0.47883	9.9176	2.6090	3.8012	0.34570	11.603
2.38	0.52528	6.4418	2.0213	3.1869	0.54862	7.7741	2.95	0.47821	9.986	2.6206	3.8106	0.34275	11.679
2.39	0.52419	6.4974	2.0308	3.1994	0.54438	7.8354	2.96	0.47760	10.055	2.6322	3.8200	0.33982	11.755
2.40	0.52312	6.5533	2.0403	3.2119	0.54015	7.8969	2.97	0.47699	10.124	2.6438	3.8294	0.33692	11.831
2.41	0.52206	6.6094	2.0499	3.2243	0.53594	7.9587	2.98	0.47638	10.194	2.6555	3.8387	0.33404	11.907
2.42	0.52100	6.6658	2.0595	3.2366	0.53175	8.0207	2.99	0.47578	10.263	2.6672	3.8479	0.33118	11.984
2.43	0.51996	6.7224	2.0691	3.2489	0.52758	8.0830	3.00	0.47519	10.333	2.6790	3.8571	0.32834	12.061
2.44	0.51894	6.7792	2.0788	3.2611	0.52344	8.1455	3.50	0.45115	14.125	3.3150	4.2608	0.21295	16.242
2.45	0.51792	6.8362	2.0885	3.2733	0.51932	8.2083	4.00	0.43496	18.500	4.0469	4.5714	0.13876	21.068
2.46	0.51691	6.8935	2.0982	3.2854	0.51521	8.2714	4.50	0.42355	23.458	4.8751	4.8119	0.09170	26.539
2.47	0.51592	6.9510	2.1080	3.2975	0.51112	8.3347	5.00	0.41523	29.000	5.8000	5.0000	0.06172	32.654
2.48	0.51493	7.0088	2.1178	3.3095	0.50706	8.3983	6.00	0.40416	41.833	7.941	5.2683	0.02965	46.815
2.49	0.51395	7.0668	2.1276	3.3214	0.50303	8.4622	7.00	0.39736	57.000	10.469	5.4444	0.01535	63.552
2.50	0.51299	7.1250	2.1375	3.3333	0.49902	8.5262	8.00	0.39289	74.500	13.387	5.5652	0.00849	82.865
2.51	0.51204	7.1834	2.1474	3.3451	0.49502	8.5904	9.00	0.38980	94.333	16.693	5.6512	0.00496	104.753
2.52	0.51109	7.2421	2.1574	3.3569	0.49104	8.6549	10.00	0.38757	116.50	20.388	5.7143	0.00304	129.217
2.53	0.51015	7.3010	2.1674	3.3686	0.48709	8.7198	∞	0.37796	∞	∞	6.000	0	∞
2.54	0.50923	7.3602	2.1774	3.3802	0.48317	8.7850							